

## Unit 1, Lesson 11: Perpendicular Bisectors



### Benchmark

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.



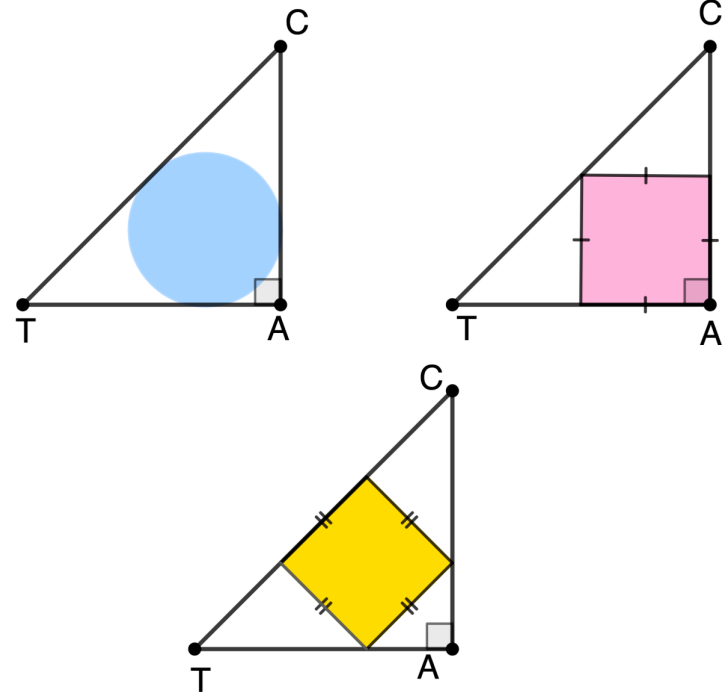
### Learning Targets

- ☐ I can construct a perpendicular bisector.
- ☐ I can use rigid transformations to show that any triangle formed by drawing two segments from the endpoints of a segment to a point on its perpendicular bisector is isosceles.
- ☐ I can use the properties of a perpendicular bisector to solve problems.



### 1.11.1 Warm-Up: Compare Areas

$\triangle CAT$  is an isosceles right triangle.



1. Which inscribed figure has the greatest area? Justify your choice.



### 1.11.2 Exploration: Constructing Perpendicular Bisectors

1. Use the steps shown to construct the perpendicular bisector of  $\overline{AB}$ .

Step 1: Set the compass point at point  $A$  and open the compass to a width approximately two-thirds the length of  $\overline{AB}$ .

Step 2: Without changing the width of the compass, draw an arc above and below the line centered at point  $A$ .

Step 3: Repeat Steps 1 and 2 to create an arc centered at point  $B$ .

Step 4: Mark the two points of intersections of the arcs as  $C$  and  $D$ . Draw a line creating  $\overline{CD}$  that intersects  $\overline{AB}$ .

Step 5: Label the point of intersection for  $\overline{CD}$  and  $\overline{AB}$  as point  $X$ .



2. Use a protractor and ruler to find each of the measurements.

| Angle        | Measurement | Segment         | Measurement |
|--------------|-------------|-----------------|-------------|
| $\angle AXD$ |             | $\overline{AB}$ |             |
|              |             | $\overline{AX}$ |             |
| $\angle AXC$ |             | $\overline{BX}$ |             |
|              |             | $\overline{CB}$ |             |
| $\angle BXD$ |             | $\overline{DB}$ |             |
|              |             | $\overline{CA}$ |             |
| $\angle BXC$ |             | $\overline{DA}$ |             |

3. Complete the statements.

The angle formed by the line of intersection of  $\overline{AB}$  and  $\overline{CD}$  is

- ☐  $85^\circ$   
☐  $90^\circ$   
☐  $105^\circ$

. The perpendicular bisector  $\overline{CD}$  divides

$\overline{AB}$  into

- ☐ 2  
☐ 3  
☐ 4

congruent parts.

4. On the perpendicular bisector  $\overline{CD}$ , add three points labeled  $Q$ ,  $R$ , and  $S$  in different places.

5. Using a ruler, measure the length of each line segment.

| Segment         | Length |
|-----------------|--------|
| $\overline{QA}$ |        |
| $\overline{QB}$ |        |
| $\overline{RA}$ |        |
| $\overline{RB}$ |        |
| $\overline{SA}$ |        |
| $\overline{SB}$ |        |

6. Write a conjecture about the endpoints of a line segment and any point on its perpendicular bisector.

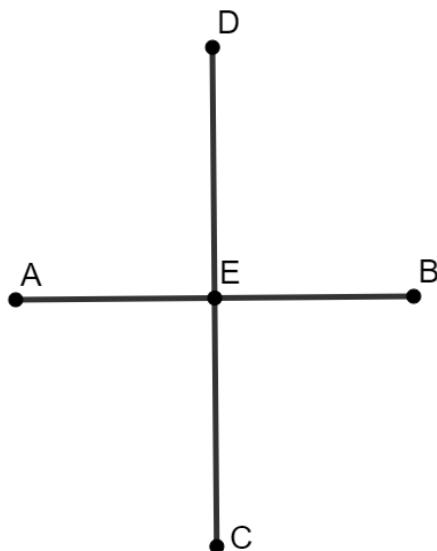
7. Share your conjecture with a partner. Summarize your partner's conjecture and have them initial that your summary is correct.

| Partner's Conjecture | Partner's Initials |
|----------------------|--------------------|
|                      |                    |



### 1.11.3 Activity: Perpendicular Bisectors and Triangles

- Copy  $\overline{AB}$  and its perpendicular bisector  $\overline{CD}$  on patty paper.



- Draw  $\overline{AD}$  and  $\overline{BD}$ .
- Make a conjecture about  $\triangle ADE$  and  $\triangle BDE$ .



A **right triangle** is a triangle containing an interior right angle.

- Reflect  $\triangle ADE$  over  $\overline{CD}$ .
- Discuss with your partner what the reflection shows is true about  $\triangle ADE$  and  $\triangle BDE$ . Summarize your discussion.



An **isosceles triangle** contains at least two equal length sides and two equal interior angle measures.

- Complete the statements.

In  $\triangle ABD$ ,  $\angle DAB$  is congruent to  $\begin{matrix} \circ \angle ADB \\ \circ \angle DBA \end{matrix}$  because a

reflection is a rigid motion that preserves angle measure.

Since  $m\angle DAB$  is equal to  $\begin{matrix} \circ m\angle ADB, \\ \circ m\angle DBA, \end{matrix}$  and  $\begin{matrix} \circ AD = AB, \\ \circ AD = BD, \\ \circ AE = BE, \\ \circ \text{None of these} \end{matrix}$

by definition  $\triangle ABD$  is a(n)  $\begin{matrix} \circ \text{scalene} \\ \circ \text{isosceles} \\ \circ \text{equilateral} \end{matrix}$  triangle.

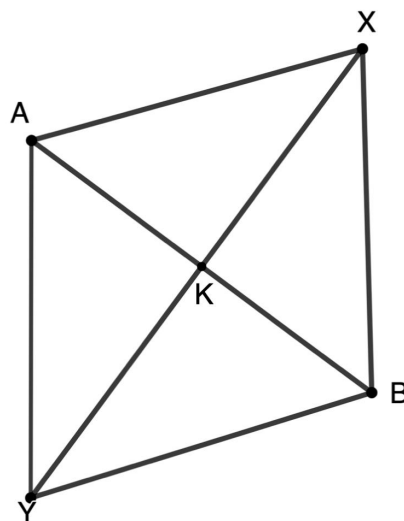


### 1.11.4 Bring It Together: Perpendicular Bisectors and Triangles



If a point is on the perpendicular bisector of a line segment, then it is equidistant from the endpoints of the line segment.

In the diagram,  $\overline{XY}$  is the perpendicular bisector of  $\overline{AB}$ ,  $XA=7.7$  millimeters (mm),  $YB=6.85$  mm, and  $AB=7.7$  mm. The figure is not drawn to scale.



1. Label all segment measurements from the given information on the image. Then, label all other segments based on the relationship of a line segment and its perpendicular bisector.
2. Given  $\overline{XY}$  is the perpendicular bisector of  $\overline{AB}$ , label all of the congruent angles.

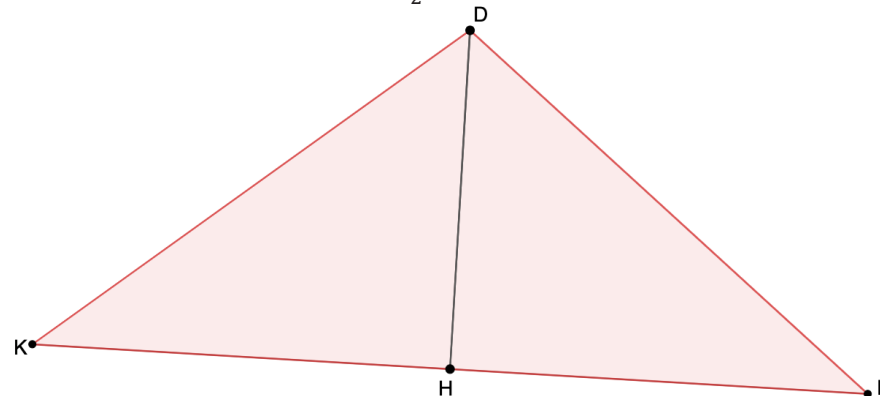


The triangle formed by a line segment and two segments drawn from the endpoints of the segment to a point on the perpendicular bisector of the segment is an isosceles triangle.



### 1.11.5 Guided Practice: Perpendicular Bisectors

1.  $\overline{DH}$  is the perpendicular bisector of  $\overline{KP}$ , where  $KD = -x - 1$  and  $PD = -\frac{x}{2} + 19$ .

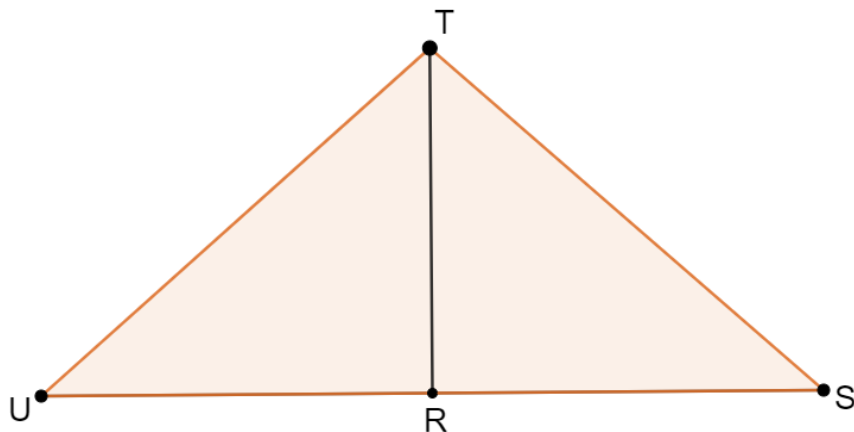


- a. What is the value of  $x$ ?
- b. What is the length of  $\overline{KD}$ ?

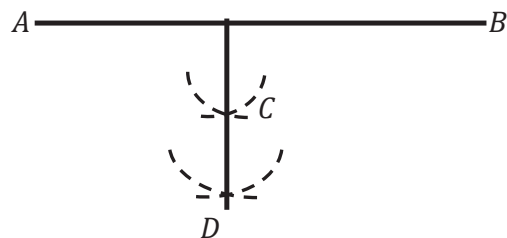


### 1.11.6 Your Turn!

1.  $\overline{RT}$  is the perpendicular bisector of  $\overline{US}$ .  $UT = 8y - 12$  and  $ST = 3y + 23$ . Find the value of  $y$ .



2. Describe the error that was made in the construction of a perpendicular bisector.



### 1.11.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.